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🌐 Portfolio [↗](#)

🐙 GitHub [↗](#)

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👤 Hugging Face [↗](#)

🏆 Kaggle [↗](#)

Summary

AI/ML Engineer who designs and ships end-to-end AI systems: a multi-agent RAG platform with a custom two-stage recommendation engine, an Arabic NLP knowledge-graph and retrieval pipeline behind a publicly released dataset, and a multimodal speech-vision-language pipeline scoring 79% on an ArabicNLP 2026 shared task — roughly double the organizers' baseline. Works across the full ML lifecycle (ingestion, feature engineering, evaluation, deployment design) on a software engineering foundation in Python, Django, FastAPI, and PostgreSQL. Completing a 9-month AI & Data Science diploma on an MCIT scholarship; seeking an AI Engineer, Machine Learning Engineer, or Data Scientist role.

Skills

Languages: Python, SQL, C • **Spoken:** Arabic (Native), English (Professional)

Machine Learning: Scikit-learn, XGBoost, Supervised & Unsupervised Learning, Feature Engineering, Hyperparameter Tuning (Grid Search), Model Evaluation & Benchmarking

Deep Learning: PyTorch, TensorFlow, Keras, Neural Networks, Computer Vision

LLM & RAG: Retrieval-Augmented Generation, Vector Databases (pgvector), Embeddings (Sentence-Transformers), Multi-Agent Systems, Prompt Engineering, JSON-Schema-Constrained Decoding, Knowledge Graphs (NetworkX, Wikidata Entity Linking), Vision-Language Models (Qwen-VL), LLM Serving (Groq API, Ollama)

NLP & Speech: CAMEL Tools (Arabic NLP), NLTK, Named Entity Recognition, Topic Modeling (BERTopic), Arabic Text Normalization, Speech-to-Text (Whisper / faster-whisper)

Data Science: Pandas, NumPy, Matplotlib, Plotly, Exploratory Data Analysis, Statistical Evaluation, Time-Series Analysis

Backend & Data Engineering: Django, FastAPI, Flask, REST APIs, Celery, SQLAlchemy, Scrapy, ETL Pipelines, Min-Hash/LSH Deduplication, PostgreSQL, pgvector, Redis

MLOps & Tooling: Offline Evaluation Harnesses (NDCG@10, MAP, Recall@k, F1), Ablation Testing, Schema-Validated Pipelines (Pydantic), Docker & Docker Compose, Git, GitHub Actions, Linux, Streamlit

Projects

AI Compass — Multi-Agent AI News & Recommendation Platform

May 2026 – Present

- Led a cross-functional team of five as Team Lead and primary technical contributor, owning system architecture, the AI pipeline, backend/database design, and deployment for a three-service platform (Python/SQLAlchemy pipeline, Django 5.2, Next.js 16) ingesting 7 sources including arXiv, GitHub, Hugging Face and YouTube.
- Designed a 5-agent enrichment and RAG pipeline (local Sentence-Transformer embeddings, PostgreSQL/pgvector, Groq-served LLMs) powering semantic search and a streaming RAG chat assistant.
- Built a deterministic two-stage recommendation engine — pgvector-similarity and rule-based candidate generation, multi-factor weighted scoring, and MMR diversification — validated with a custom offline NDCG@10/MAP harness with shadow-mode comparison.
- Owned async infrastructure (Celery + Redis, 3 dedicated queues, 6-job scheduler) and a containerized Docker Compose deployment with Caddy-routed TLS; diagnosed and fixed duplicate task redelivery caused by an 88-minute run exceeding the broker visibility timeout.

Palestinian Cultural Knowledge Platform [↗](#) — Arabic NLP, RAG & Knowledge Graph

Jun – Aug 2026

- Primary technical contributor on a 2-person research project turning a quality-scored, deduplicated Arabic/English corpus from 5 sources into structured, queryable knowledge.
- Designed the ingestion pipeline (schema-validated records, composite quality scoring, persistent MinHash/LSH deduplication) and led Arabic NER (CAMEL Tools, F1

0.47) and RAG (embeddings → pgvector → locally-served LLM, Recall@5 0.93 over 1,282 indexed passages).

- Built the knowledge-graph pipeline (entity canonicalization, Wikidata entity linking, LLM relation extraction) producing 13,063 canonicalized entities and a 187-edge graph, plus bias and topic analysis (WEAT, BERTopic) served through a FastAPI REST API and a Streamlit dashboard.
- Published a bilingual Arabic/English dataset — 882 documents, ~890K words — on [Hugging Face Datasets](#).

AynVQA-MSA — Arabic Spoken Visual Question Answering

Jul – Aug 2026

- Built an end-to-end multimodal pipeline (audio, vision & language) for the **ImageEval 2026 / ArabicNLP** shared task: Whisper ASR → JSON-schema-constrained LLM parsing of spoken questions and options → conditional repair escalation → vision-language-model answer selection.
- Reached **79.00% accuracy (395/500)** on the official development split — nearly double the organizers' MSA baseline (39.8%) — lifting a 74.80% baseline with a 3-step repair-escalation stage, and replacing fragile regex parsing with schema-constrained decoding; chain-of-thought and few-shot prompting were ablated and shipped off by default on measured results.

Fitness Activity Tracker — ML-Powered Exercise Classification

Mar – Apr 2026

- 3-person team project (Digilians AI & Data Science Diploma) classifying barbell exercises and counting repetitions from wrist-worn accelerometer/gyroscope data; owned preprocessing, feature engineering (Chauvenet outlier removal, PCA, FFT frequency-domain features, k-means clustering), and the evaluation pipeline — compared 5 classifiers across 5 feature sets via grid search, with Random Forest reaching ~99% held-out test accuracy.

Work Experience

Freelance Backend Developer — StockGo (Qatar)

Nov 2024 – Apr 2025

- Built and deployed a Django REST API and business-analytics dashboard for a live commercial inventory-tracking product (thestockgo.com), covering data modeling, API design, and reporting queries over PostgreSQL.

Backend Developer Intern — Venture Appital

Oct 2024 – Feb 2025

- Built modular RESTful microservices in Flask with full unit-test coverage.

Full Stack Developer Intern — Nftrtix

Mar 2025 – Apr 2025

- Built an internal Django/JavaScript web application to streamline internal operations.

Education

Faculty of Engineering, Alexandria University

Sep 2019 – Jun 2024

BS in Communication and Electronics Engineering — CGPA 3.13 / 4.0

- Graduation Project: **Advanced Driver Assistance Systems (ADAS)** — Grade A+, ranked 5th of ~60 projects (Value competition). Team Lead over a 5-repository, ~800 KB bare-metal C system; owned the software architecture and computer-vision track (lane keeping, automatic emergency braking, sign detection). Mentored by Valeo, sponsored by ITIDA's Egypt Makes Electronics (EME) program.

Certifications & Training

- **AI & Data Science Diploma** — Digilians (MCIT / Egyptian Military Academy scholarship) Dec 2025 – Sep 2026
- **IBM AI Engineering Professional Certificate** — IBM / Coursera Jun 2026
13 courses: deep learning with PyTorch, Keras & TensorFlow; transformers, fine-tuning, RAG with LangChain
- **Machine Learning Specialization** — Stanford & DeepLearning.AI (Andrew Ng) May 2026
- **Generative AI for Data Science with Copilot** — Microsoft / Coursera Apr 2026
- **Microsoft Data Engineer (159 hours)** — Digital Egypt Pioneers Initiative (DEPI) Apr – Oct 2024
- **AWS Certified Machine Learning Engineer – Associate (MLA-C01)** In Progress